BRANDON SANTOR

Geospatial researcher | Software developer | U.S. Marine Corps veteran

emp1241.github.io

Geospatial researcher and software developer building scientific tools from the data pipeline to the interface. My work spans tree-ring and climate analysis, deep-ocean WebGIS, reconstructed Martian terrain, spatial evidence systems, and independent GNSS research. Texas A&M GIST undergraduate and U.S. Marine Corps veteran.

RESEARCH & PROFESSIONAL EXPERIENCE

Research Assistant & Software Developer | Torbenson Lab

March 2026-present | Paid appointment since June 2026

- Independently created TRAE in response to lab research needs; the prototype led to a paid appointment for continued development and research support.
- Built reviewed scientific imports, linked maps and graphs, climate matching, saved sessions, and exports with provenance. Added PRISM, NetCDF, ROXAS-output preparation, and WalDendro cleaning workflows.
- Applied TRAE to derive, clean, organize, and export data for a co-authored manuscript in preparation.

Research Lead & GIS Developer | NWSW

Fall 2026-present | Texas A&M GIS Practicum

- Lead cross-team evidence synthesis, source reconciliation, reusable boundary preparation, geographic joins, and evidence-matrix integration.
- Used my Research Architect framework to generate the project's coordination system, with versioned context, SHA-256 file checks, and reviewed contributor handoffs.
- Independently built a separate spatial evidence application using FastAPI, PostGIS, TypeScript, and Leaflet.

Independent Esri GIS Technical Consultant & Instructor

February-May 2026

- Audited seven graduate GIS modules and rebuilt six workflows into 189 pages of illustrated technical manuals.
- Corrected ArcGIS software, projection, raster alignment, geodatabase schema, join, licensing, and desktop-to-web failures.
- Built quality-control checkpoints and taught reproducible workflows that enabled the client to execute corrected analyses independently.

High School Baseball Coach

2013-2020 | Paid program coach

- Led coaching for nearly 100 student-athletes, developing baseball skills, teamwork, and discipline.
- Coordinated conditioning, equipment, and maintenance for the program.
- Mentored student-athletes and emphasized education throughout their high school experience.

EDUCATION

Texas A&M University
B.S. Geographic Information Science & Technology
Expected May 2027
Institutional GPA: 4.0 / 4.0

ITT Technical Institute
A.S. Computer Networking | 2007

HONORS & SCHOLARSHIPS

Second Place | TxGIS Day Undergraduate Student Research Presentation Competition
2025 | Health Geography GIS and statistical analysis

Galloway '29 Aggie Veteran Freedom Scholarship II
Two-time recipient | 2025-2027

Dennis W. Hughes '79 & Melinda Hood Hughes '81 Endowed Scholarship in Geography
Two-time recipient | 2025-2027

Richard A. & Mollie B. Williford Dean's Scholarship
College of Arts & Sciences | 2026-2027

RESEARCH & PROFESSIONAL EXPERIENCE - CONTINUED

U.S. Marine Corps | Infantryman, Acting Platoon Sergeant & Logistics NCO

2011-2013

- Managed logistical support, inventory, and resource allocation for a company-sized element of approximately 350 Marines.
- Coordinated personnel and mission-critical field operations under strict safety, accountability, and risk-management requirements.
- Applied disciplined decision-making, communication, and logistical problem-solving in high-pressure field environments.

TEKsystems | Network & Computer Systems Contractor

2007-2011

- Executed enterprise hardware and software deployments involving 100+ devices per project during nationwide infrastructure upgrades.
- Independently diagnosed complex computer and network systems issues across changing client environments.
- Installed, configured, and supported systems while coordinating with local and remote technical teams.

SELECTED SOFTWARE & RESEARCH

Mars Explorer | Team lead | Database & 3D systems

- Lead a team of five; design and implement the 41-table spatial database, source pipelines, API, and custom Cesium terrain system. Transform orbital elevation and imagery into textured meshes and bounded exploration.

Thermal Vent Atlas Explorer | Independent creator & developer

- Delivered an online/offline research WebGIS to Sylvan Lab for review and internal use, connecting 742 atlas records with bathymetry, tectonic boundaries, geomicrobiology context, comparisons, sessions, and scoped exports.

COLE | Cost Of Living Explorer | Project lead | Feature integration

- Lead design and feature add-on integration for household-driven county search. Combine county household-cost estimates, climate, geographic lookup, and optional hazard context with visible assumptions and three-county comparison.

PhalAInx | Independent software studio | Creator & principal designer

- Created Research Architect, which builds individual AI agent kits for human teams with shared methods, context, review, and handoffs. Built High Command v0.5.6, a React/TypeScript prototype for a personal multi-model workspace; alpha testing. Selected Tauri/Rust/SQLite desktop integration and model coordination remain in development.

The Deception Gap | Smartphone GNSS | Independent continued research

- Expanded a class project into a 781-reading study across three smartphones and five crowd categories; led continued collection, statistical analysis, interpretation, and poster development.

NASA SUITS Challenge Proposal | Team lead & primary system designer | 2021-2022

- Led conceptual HoloLens 2 EVA-interface design with local positioning, navigation, terrain/hazard overlays, and geological annotation; co-authored technical documentation submitted to NASA for review.

TECHNICAL TOOLKIT & RESEARCH COMMUNICATION

Python; JavaScript / TypeScript; PostgreSQL / PostGIS; FastAPI / Flask; Cesium / Leaflet; NumPy / pandas / Rasterio; ArcGIS Pro / Field Maps; QGIS / GDAL; React / Vite; Scientific import / export; GNSS field analysis; Research provenance; JSON Schema; SHA-256 manifests.

Co-authored manuscript in preparation with Torbenson Lab. Scientific data preparation, workflow development, and research presentation.